

A Compact 0.3-V Class AB Bulk-Driven OTA

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Abstract—In this article, a new solution for an ultralow-voltage (ULV) ultralow-power (ULP) operational transconductance amplifier (OTA) is presented. Thanks to the combination of a low-voltage bulk-driven nontailed differential stage with the multipath Miller zero compensation technique, a simple class AB power-efficient ULV structure has been obtained, which can operate from supply voltages less than the threshold voltages of the employed MOS transistors, while offering rail-to-rail input common-mode range at the same time. The proposed OTA was fabricated using the 180-nm CMOS process from Taiwan Semiconductor Manufacturing Company (TSMC) and can operate from V_{DD} ranging from 0.3 to 0.5 V. The 0.3-V version dissipates only 12.6 nW of power while showing a 64.7-dB voltage gain at 1-Hz, 2.96-kHz gain-bandwidth product, and a 4.15-V/ms average slew-rate at 30-pF load capacitance. The measured results agree well with simulations.

Index Terms—Bulk-driven (BD), low power, low voltage, operational amplifier, operational transconductance amplifier (OTA).

I. INTRODUCTION

IN RECENT years, there has been an increased interest in the design of ultralow-voltage (ULV) and ultralow-power (ULP) analog circuits. This new trend is mainly motivated by the increased demand for power-efficient solutions used in medical implantable or wearable devices, wireless sensor nodes, and other similar applications. Lower supply voltage allows decreasing the dissipated power of both, analog as well as a digital portion of the above-mentioned systems.

One of the most widely used analog building blocks in CMOS circuits is the operational transconductance amplifier (OTA). A number of ULV OTAs have been proposed in the literature in recent years [1]–[17]. It is worth noting that more and more of these circuits can operate even from deep sub-0.5-V supply [10], [13], [16].

The sub-0.5-V OTAs are generally based on subthreshold biased bulk-driven (BD) transistors. Even though the BD transistors show a number of disadvantages (lower transconductance, lower voltage gain, larger input noise, and offset), this technique seems to be still attractive in such cases when the supply voltage (V_{DD}) is very low (comparable with the thresh-

old voltage V_{TH} of MOS transistors) and a rail-to-rail input common-mode range (ICMR) is required at the same time.

The bulk of the conventional gate-driven (GD) MOS differential pair can also be used as a control terminal to lower the threshold voltage [3] and provide common-mode control, ultimately removing the need for the tail current generator, and increasing the ICMR [17]. Moreover, as shown recently, forward bulk biasing increases the intrinsic voltage gain of short-channel MOS transistors in the ULV environment [18].

Several interesting solutions for deep sub-0.5-V BD OTAs have recently been proposed in the literature. In [10], a 0.25-V OTA based on low- V_{TH} transistors and a gain boosting circuit has been described. In [13], a sub-0.5-V three-stage OTA with pseudodifferential input stage has been proposed. The circuit was implemented in a 65-nm process and showed a gain-bandwidth product (GBW) in the megahertz range. In [16], a family of class AB 0.3-V OTAs, fabricated using a 180-nm CMOS process, has been shown. Thanks to the application of nontailed differential input stages and a class AB output stage, the circuits can operate with very low relative (V_{DD}/V_{TH}) supply voltage while offering very competitive performance in terms of GBW and slew rate (SR).

In this article, a new solution for a deep sub-0.5-V OTA is described. Thanks to the combination of the BD nontailed differential stage [19] with the multipath Miller zero cancellation compensation (MMZCC) technique [20], a simple class AB power-efficient ULV structure has been obtained with rail-to-rail ICMR. The resulting structure of OTA is compact, easy to design, and does not require any auxiliary circuits (like replica bias) while offering better power efficiency and common-mode rejection ratio (CMRR) than other sub-0.5-V OTAs in the literature.

The rest of the article is organized as follows. In Section II, the principle of operation of the proposed OTA is described. The details of implementation are discussed in Section III, while the experimental and simulation results are provided in Section IV. Finally, the article is concluded in Section V.

II. CIRCUIT DESCRIPTION

The schematic of the proposed amplifier is shown in Fig. 1. The circuit can be considered as a two-stage Miller OTA with an additional feedforward path (M_5 – M_7) used to compensate for the right-half-plane (RHP) zero introduced by the compensation capacitance C_C . The input stage (M_1 – M_4) is based on the idea of a nontailed BD differential pair with gain boosting [19]. The circuit can operate with very low supply voltage ($V_{DDmin} = 2 V_{DSsat}$) while offering good CMRR, power supply rejection ratio (PSRR), and rail-to-rail ICMR at

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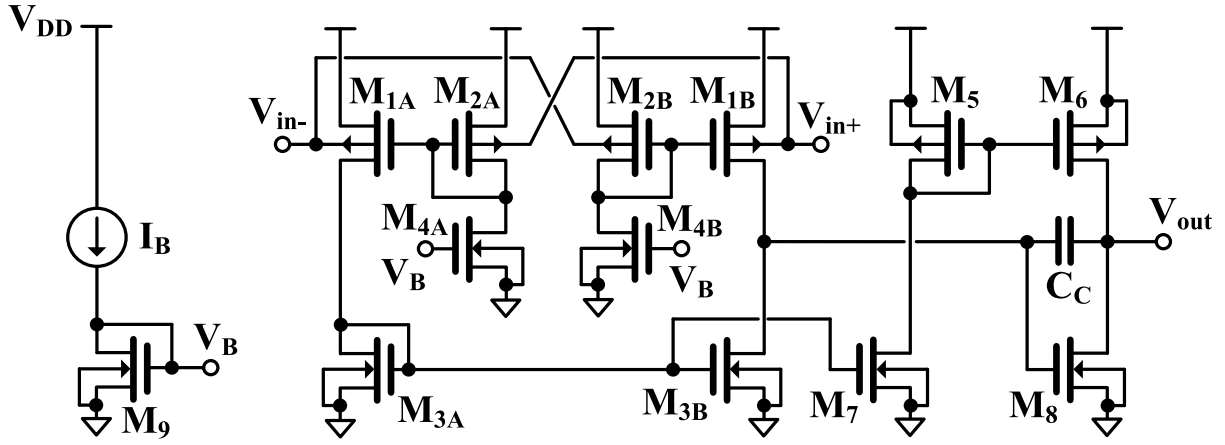


Fig. 1. CMOS schematic of the proposed OTA.

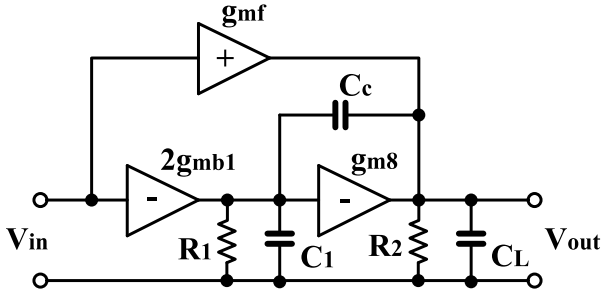


Fig. 2. Block diagram of OTA.

the same time [16], [19], [21]. In addition, the circuit operates in class AB, which increases its SR performance. It is worth noting that its noise properties and the input-referred offset are not worse than achieved for a conventional BD differential pair (see the Appendix).

The second gain stage of the OTA is formed by the transistor M_8 , loaded by the transistor M_6 . The capacitor C_C is used for frequency compensation. As it is well known, the compensation capacitor introduces a parasitic RHP zero, which reduces the phase margin (PM) and, consequently, limits the GBW product of a two-stage Miller OTA. There are several techniques to mitigate or even completely remove the impact of the RHP zero. In the proposed design, the MMZCC technique [20] has been applied. The block diagram of the OTA with MMZCC is shown in Fig. 2, where $2g_{mb1}$ stands for the transconductance of the input stage [19], g_{m8} is the transconductance of the output stage, and g_{mf} is the transconductance of the feedforward path, which compensates for the direct feedthrough effect associated with C_C . In the proposed design, this path is created by the input stage and the transistors M_5 – M_7 . The resistances R_1 and R_2 represent the output resistances of the input and output stages, respectively, while C_1 and C_L represent their load capacitances.

A. Small-Signal Performance

The transfer function of the MMZCC structure shown in Fig. 2 is given by

$$\frac{V_{out}}{V_{in}} \cong \frac{2g_{mb1}g_{m8}R_1R_2\left[1 + s\frac{(g_{mf}-2g_{mb1})C_C}{2g_{mb1}g_{m8}}\right]}{(1 + sg_{m8}R_2R_1C_C)(1 + sC_L/g_{m8})}. \quad (1)$$

Thus, the transfer function (1) has two left-half-plane (LHP) poles and one zero as follows:

$$p_1 \cong -\frac{1}{g_{m8}R_2R_1C_C} \quad (2)$$

$$p_2 \cong -\frac{g_{m8}}{C_L} \quad (3)$$

$$z_1 \cong -\frac{2g_{mb1}g_{m8}}{(g_{mf} - 2g_{mb1})C_C}. \quad (4)$$

Note that the zero z_1 can be LHP if $g_{mf} > 2g_{mb1}$. It is also possible to cancel p_2 with z_1 if the following condition is met:

$$g_{mf} = 2\left(1 + \frac{C_L}{C_C}\right)g_{mb1}. \quad (5)$$

The transconductance g_{mf} for the considered structure can be expressed as

$$g_{mf} = g_{mb1}\frac{g_{m7}}{g_{m3}} \cdot \frac{g_{m6}}{g_{m5}}. \quad (6)$$

Thus, from (5) and (6), assuming that all transistors operate in the subthreshold region where their transconductances are proportional to the biasing currents, we obtain the following condition for pole-zero cancellation:

$$\frac{I_{D6}}{I_{D3}} = 2\left(1 + \frac{C_L}{C_C}\right). \quad (7)$$

When the above condition is met, the transfer function (1) shows one dominant pole p_1 only. It is worth noting that meeting this condition in ULP circuits may be difficult because a relatively large quiescent current of the output stage (I_{D6}) is generally required. The I_{D6}/I_{D3} ratio may be lowered with increasing C_C ; however, this leads to lower GBW, SR, and increased silicon area. Thus, the above compensation is useful for small loading capacitances (i.e., small C_L/C_C ratios achieved for relatively small C_C).

In the proposed design, a current push–pull version of the feedforward path g_{mf} was applied [22], which improves the driving capability of the output stage. The drain current of M_3 is mirrored to the output through the current mirrors M_3 – M_7 and M_5 – M_6 , thus increasing the current sourced to the load. On the other hand, M_8 provides the sinking current ability. Consequently, a push–pull amplifier is created in a very simple manner.

The open-loop dc voltage gain of the OTA is not significantly affected by the feed-forward path and can be expressed as

$$A_{vo} \cong \frac{2g_{mb1}}{(g_{ds1} + g_{ds3})} \cdot \frac{g_{m8}}{(g_{ds6} + g_{ds8})} \quad (8)$$

while the GBW product is

$$GBW \cong \frac{2g_{mb1}}{C_C}. \quad (9)$$

In case of ideal symmetry of the input stage, the dc CMRR may be expressed as

$$CMRR \cong \frac{2g_{mb1}}{(g_{ds2} + g_{ds4})} \cdot \frac{g_{m3}}{(g_{ds1} + g_{ds3})} \cdot \frac{g_{m5}}{(g_{ds5} + g_{ds7})}. \quad (10)$$

Therefore, it is of the order of a voltage gain of a three-stage amplifier and can approach very large values, which is an advantage of this (nontailed) OTA.

The dc voltage gain from V_{DD} to output is approximately equal to unity; hence, the PSRR is approximately equal to the open-loop voltage gain of the OTA, which is given by (8).

B. Slew-Rate

The negative SR of the considered OTA may be expressed as

$$SR_- = \min \left(\frac{I_{omax}}{C_C}, \frac{I_{D8max}}{C_C + C_L} \right) \quad (11)$$

where I_{D8max} is the maximum current sinking by M_8 and I_{omax} is the maximum output current of the input stage, which for $I_{D1} = I_B$ at the operating point is given by

$$I_{omax} \cong 2I_B \sinh \left(\frac{\eta V_{inm}}{n_p U_T} \right) \quad (12)$$

where V_{inm} is the assumed value of the input differential voltage, which can approach V_{DD} , n_p is the subthreshold slope, U_T is the thermal potential, and $\eta = g_{mb}/g_m$. Generally, SR_- is limited by I_{omax} and C_C . It is worth noting, that due to the expansive-type of the transfer characteristic of the input stage, the current I_{omax} is not limited by a tail current source and can exceed the total biasing current of the input stage, which improves the SR performance [16].

Neglecting the parasitic time constants associated with the internal nodes of OTA, the SR_+ may be expressed as

$$SR_+ \cong \min \left(\frac{I_{omax}}{C_C}, \frac{I_{omax}}{2C_L} \cdot \frac{I_{D6}}{I_{D3}} \right). \quad (13)$$

Assuming that the SR is constrained by C_C , and the SR_+ by C_L , the SR_+/SR_- ratio is

$$\frac{SR_+}{SR_-} = \frac{C_C}{2C_L} \cdot \frac{I_{D6}}{I_{D3}}. \quad (14)$$

Thus, SR_+ is equal to SR when $I_{D6}/I_{D3} = 2 C_L/C_C$.

TABLE I
TRANSISTOR ASPECT RATIOS

Device	W/L [$\mu\text{m}/\mu\text{m}$]	Device	W/L [$\mu\text{m}/\mu\text{m}$]
$M_{1A,B}, M_{2A,B}$	15/1	M_6	5x90/0.4
$M_{4A,B}$	15/0.5	M_7	2x20.6/1
$M_{3A,B}$	20.6/1	M_8	10x20.6/1
M_5	90/0.4	M_9	15/0.5

C. Noise and Offset

The input-referred noise and offset of the OTA are both determined by properties of its input stage. Detailed formulas and comparison with a conventional BD pair can be found in the Appendix. It is worth pointing out that both the input noise and offset of the considered circuit are not worse than achieved for a conventional BD pair, provided that the conditions given in the Appendix are met.

III. IMPLEMENTATION

The proposed OTA has been designed and fabricated in a 0.18- μm CMOS process from Taiwan Semiconductor Manufacturing Company (TSMC) ($V_{TH} \approx \pm 0.5$ V). The circuit was designed for $V_{DD} = 0.3$ V, assuming operation in deep subthreshold region, and with the following specifications: $GBW = 3.7$ kHz, $C_L = 20$ pF, $PM = 60^\circ$, and the dynamic range (DR) = 60 dB.

The biasing current I_B was calculated on the basis of the assumed input noise and DR [16]. It was also assumed $M_1 = M_2$ for optimal noise performance [19] and $M_9 = M_4$. This entails $I_{D1} = I_{D2} = I_B$. For the calculated value of I_B (2.5 nA), the input transconductance ($2g_{mb1}$) was 44.5 nS ($n_p = 1.35$, $\eta = 0.31$). From (9), using the assumed value of GBW, the compensation capacitance C_C was 1.9 pF. Next, the biasing current I_{D6} (I_{D8}) was chosen to be 25 nA ($I_{D6}/I_{D3} = 10$), which provides symmetrical SR and z_1 - p_2 cancellation for $C_L \approx 8 - 10$ pF (the compensation was not predicted for $C_L = 20$ pF to decrease the dissipation power). Finally, the biasing current I_{D5} (I_{D5}/I_{D6} ratio) has been chosen to be equal to 0.2 of I_{D6} , as a tradeoff between the dissipation power and the location of the parasitic pole associated with the drain/gate node of M_5 .

Once the biasing currents were set, the W and L dimensions of all transistors were adjusted during the simulation phase. The channel lengths were chosen relatively large to increase the voltage gain. Larger L in the input stage also entails the larger transistor channel areas, which allows decreasing the input-referred offset and flicker noise. The channel widths were adjusted to achieve $|V_{GS}| \approx V_{DD}/2$ at the operating point. The aspect ratios of all transistors and small-signal parameters are summarized in Tables I and II, respectively. Fig. 3 shows the microphotograph of the fabricated OTA.

IV. RESULTS AND COMPARISON

A. Experimental and Simulated Results

Because of the fact that our printed circuit board showed relatively large parasitic capacitance loading the OTA (> 20 pF),

TABLE II
SMALL-SIGNAL PARAMETERS FOR $V_{DD} = 0.3$ V, $I_B = 2.5$ nA

Parameter	Value
$2g_{mb1}$	44.5 nA/V
g_{m8}	739 nA/V
g_{mf}	220 nA/V
C_C	1.9 pF

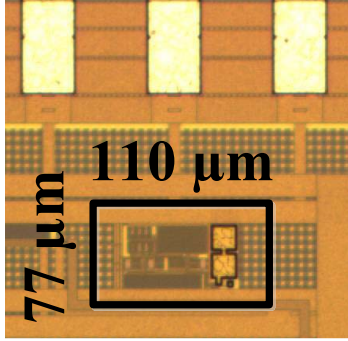


Fig. 3. Microphotograph of OTA.

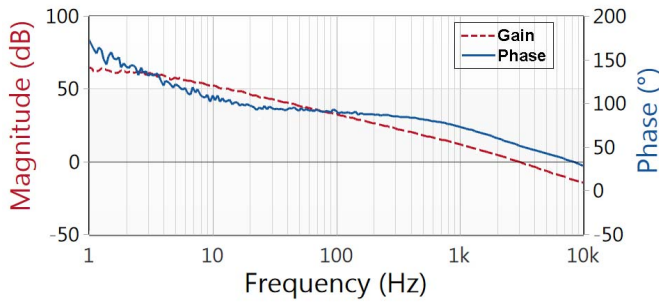


Fig. 4. Frequency characteristics of OTA. $V_{DD} = 0.3$ V, $I_B = 2.5$ nA, and $C_L = 30$ pF.

the circuit was measured with $C_L = 30$ pF. This capacitance was precisely adjusted and measured with the E498A LCR meter from Agilent.

Fig. 4 shows the open-loop magnitude and phase responses of the OTA, measured with a Bode 100 vector analyzer within 1 Hz–10 kHz. The characteristics prove proper operation of OTA, even though the GBW product and PM were slightly affected by the increased value of C_L .

Figs. 5 and 6 show the main small-signal parameters (voltage gain at 1 Hz, GBW, and PM) as a function of the input common-mode voltage (V_{icm}). The voltage gain exceeds 52 dB for V_{icm} ranging from 50 to 250 mV, i.e., 50 mV from each supply rail, with less than 6.5% variations of GBW in this range and proper PM. Even for a V_{icm} of 20 mV from each supply rail, the voltage gain exceeds 33 dB.

Fig. 7 shows the step response of the OTA in a voltage follower (VF) configuration with a 250-mV_{pp} input step. The SR_+ (1.9 V/ms) was approximately three times lower than

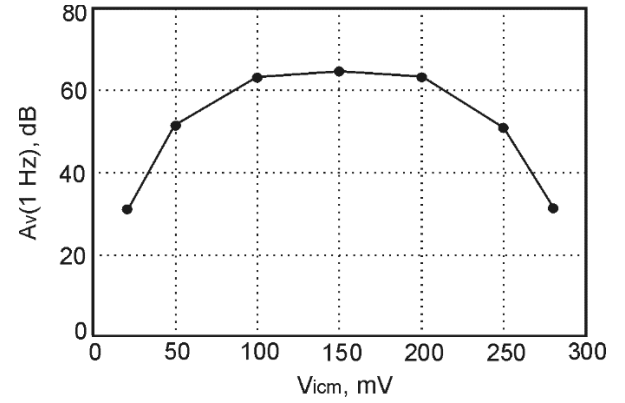


Fig. 5. Voltage gain of OTA at 1 Hz against the input common-mode voltage. $V_{DD} = 0.3$ V, $I_B = 2.5$ nA, and $C_L = 30$ pF.

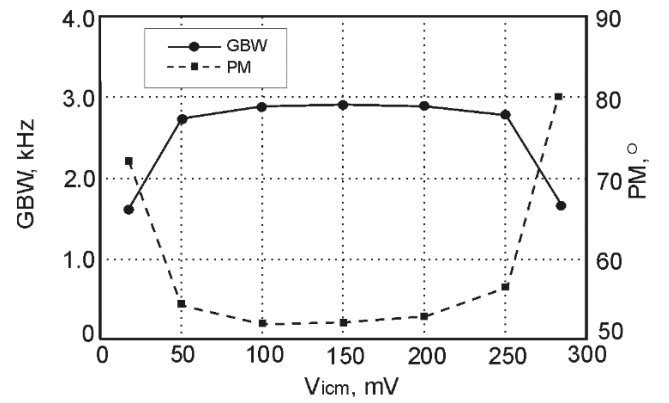


Fig. 6. GBW and PM of OTA against the input common-mode voltage. $V_{DD} = 0.3$ V, $I_B = 2.5$ nA, and $C_L = 30$ pF.

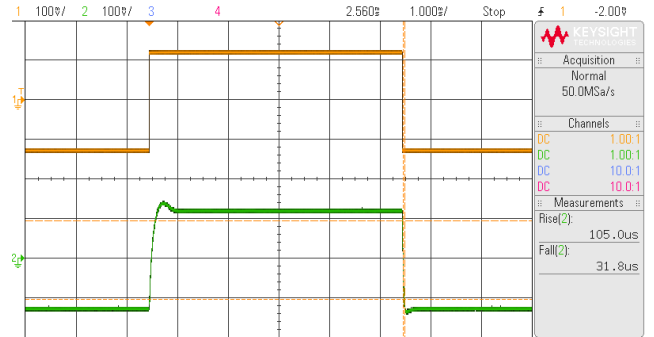


Fig. 7. Step response of OTA. $V_{DD} = 0.3$ V, $I_B = 2.5$ nA, $C_L = 30$ pF, V_{in} (yellow), V_{out} (green), horizontal 1 ms/div, and vertical 100 mV/div.

SR_- (6.4 V/ms), which was in agreement with theoretical expectations for $C_L = 30$ pF [see (14)].

Fig. 8 shows the sine-wave response of the VF for rail-to-rail input ($f = 10$ Hz). The VF response proves a rail-to-rail operation of the proposed OTA. The total harmonic distortion (THD) was 1% for $V_{in} = 255$ mV_{pp}.

The above measurements were repeated for $V_{DD} = 0.5$ V and $I_B = 40$ nA, showing that the circuit operates well in this case as well, with measured GBW = 50.1 kHz, PM = 53.2°, $A_{vo} = 68$ dB at 1 Hz, dc CMRR = 107 dB, and rail-to-rail ICMR.

TABLE III
RESULTS OF MONTE CARLO ANALYSIS, FOR $V_{DD} = 0.3$ V,
 $I_B = 2.5$ nA, $C_L = 30$ pF, (200 RUNS)

Parameter	Mean	Std. dev.	min	max
A_{vo} @ DC, [dB]	69.0	0.0178	68.94	69.04
GBW, [kHz]	3.03	0.132	2.73	3.45
PM, [°]	54.4	0.69	52.8	56.4
t_{sett} (1%), [μ s]	494	0.212	493.85	494.75
CMRR, [dB]	128	0.346	125.9	128.7
PSRR, [dB]	68.9	0.018	68.87	68.97
I_{DD} , [nA]	42.1	1.79	37.5	47.0

TABLE IV
PERFORMANCE SUMMARY OF THE PROPOSED OTA

Parameter	Value (meas.)	Value (simulated)	
V_{DD} , [V]	0.3	0.3	
C_L , [pF]	30	30	20
DC voltage gain, [dB]	-	68.9	
Voltage gain @ 1 Hz	64.7	67.1	
GBW, [kHz]	2.96	2.94	3.34
Phase margin, [°]	52	54.7	62.4
SR+/SR-, [V/ms]	1.9/6.4	2.0/6.5	2.8/6.6
1 % Settling time, [μ s]	446	483	283
Thermal noise, [μ V/Hz ^{1/2}]	-	1.6 ^{c)}	
CMRR @ DC, [dB]	110	127.6	
PSRR @ DC, [dB]	56	62.6	
Input referred offset, [mV]	3.2 ^{a)}	3.2 ^{b)}	
Dissipation power, [nW]	12.6	12.6	
Input current (0- V_{DD}), [pA]	< 100	< 80	
Silicon area, [mm ²]	0.0085	0.0085	

a) std. dev. for 5 measured samples

b) std. dev. (MC 200 runs)

c) simulated

Figs. 9 and 10 show the histograms of dc CMRR and PSRR, based on Monte Carlo (MC) analysis. Note that CMRR remains very high (much above the values reported for other sub-0.5-V OTAs) even in the presence of a mismatch. The MC analysis has also been performed for other circuit parameters. The results summarized in Table III prove that the design is robust against transistor mismatch.

The simulated and measured results for $V_{DD} = 0.3$ V and $I_B = 2.5$ nA are summarized in Table IV. The simulated results for $C_L = 20$ pF were in good agreement with design specifications. Small differences may be attributed to the second-order effects, like the impact of g_{ds} conductances or parasitic capacitances. The results for $C_L = 30$ -pF show good compatibility between simulations and measurements. Larger differences may be observed for CMRR and PSRR, since these parameters were affected by process corners.

TABLE V
SIMULATED MAIN PERFORMANCE PARAMETERS OVER PROCESS AND TEMPERATURE VARIATIONS FOR $V_{DD} = 0.3$ V,
 $I_B = 2.5$ nA, AND $C_L = 30$ pF

Parameter (T=27 °C)	SS	SF	TT	FS	FF
P_{diss} , [nW]	12	12.7	12.6	12.5	13.2
A_{vo} , [dB]	69	67.7	68.9	69.2	67.5
GBW, [kHz]	2.87	2.92	2.94	2.95	3
Phase margin, [°]	53.9	55.4	54.7	54.2	55.6
SR+/ SR-, [V/ms]	1.8/ 5.4	1.9/ 5.6	2/ 6.5	2/ 6.9	2/ 6.6
CMRR @ DC, [dB]	106	102	127	105	102
PSRR @ DC, [dB]	59	64.3	62.6	61.8	62.7

Parameter (TT corner)	0°	27°	70°
P_{diss} , [nW]	11.7	12.6	14.3
A_{vo} , [dB]	68.6	68.9	64.4
GBW, [kHz]	2.99	2.94	2.86
Phase margin, [°]	54.3	54.7	55.8
SR+/SR-, [V/ms]	1.9/5.6	2/6.5	1.9/5.9
CMRR @ DC, [dB]	103.4	127	93.8
PSRR @ DC, [dB]	57.6	62.6	60.0

Process corners: S-slow, F-fast, T-typical, first: n-channel MOS, second p-channel MOS.

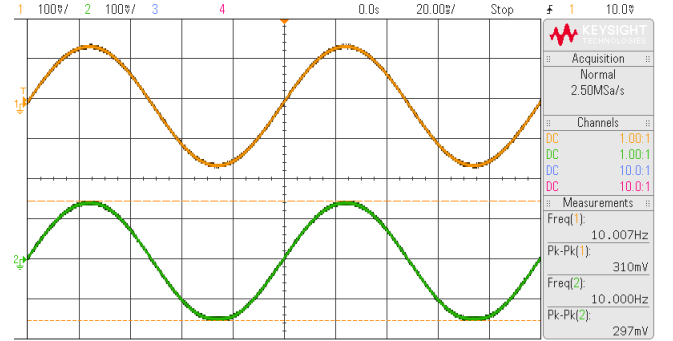


Fig. 8. Sine-wave response of OTA for rail-to-rail input. $V_{DD} = 0.3$ V, $I_B = 2.5$ nA, and $C_L = 30$ pF, V_{in} (yellow), V_{out} (green), horizontal 20 ms/div, and vertical 100 mV/div.

Table V shows the results of corner analysis, proving that the circuit is robust against process and temperature variations.

B. Performance Comparison

Table VI presents a comparison of the proposed OTA with other fabricated ULV OTAs ($V_{DD} \leq 0.7$ V), published in the literature in recent years. First of all, it is worth mentioning that the proposed circuit can operate with lower relative supply voltage (V_{DD}/V_{TH}) than other amplifiers, except [16], which

TABLE VI
PERFORMANCE COMPARISON OF SUB-0.7-V AMPLIFIERS

Parameter	This work	[16] 2018	[15] 2016	[14] 2016	[13] 2015	[11] 2014	[10] 2014	[8] 2007	[6] 2005	[6] 2005
Technology, [μm]	0.18	0.18	0.18	0.18	0.065	0.18	0.13	0.35	0.18	0.18
V_{DD} , [V]	0.3	0.3	0.7	0.5	0.35	0.5	0.25	0.6	0.5	0.5
$V_{DD}/V_{TH}^{(d)}$	0.6	0.6	1.75	1.18	1.16	1.29	1.16	1.02	1.14	1.14
C_L , [pF]	30	20	20	40	3	30	15	15	20	20
DC gain, [dB]	> 64.7	> 65.8	57.5	77	43	70	60	69.4	62	52
GBW, [kHz]	2.96	2.78	3e3	4.0	3.6e3	18	1.88	11.3	10e3	2.5e3
Phase margin, [$^\circ$]	52	61	60	56	56	55	52.5	65	-	-
SR+, [V/ms]	1.9	6.44	1.8e3	2.0	5.6e3	3.0	0.64	14	2e3	2.89e3
SR-, [V/ms]	6.4	7.80	3.8e3	2.0	5.6e3	3.0	0.77	14	2e3	2.89e3
CMRR @ DC, [dB]	110	72	19	55	46	-	-	75	74.5	78
Thermal noise, [$\mu\text{V}/\text{Hz}^{1/2}$]	1.6 ^{e)}	1.85 ^{e)}	0.10	-	-	0.31	3.3	0.29	0.070	0.080
Power, [nW]	12.6	15.4	25.4e3	70	17e3	75	18	548	74e3	110e3
Operation mode	BD	BD	BD	GD	BD	GD	BD	BD	GD	BD
ICMR/ V_{DD} , [%]	100	100	79	100	100	60	100	100	20	100
FOM ₁ , [MHz $\cdot$ pF/mA]	2114	1083	1650	1143	222	3600	392	181	1351	227
FOM ₂ , [V $\cdot$ pF/($\mu\text{s} \cdot \text{mA}$)]	2964	2790	1540	571	346	680	146	230	270	263
FOM ₃ , [MHz $\cdot$ pF/mW]	7047	3610	2357	2286	634	7200	1568	302	2702	455
FOM ₄ , [V $\cdot$ pF/($\mu\text{s} \cdot \text{mW}$)]	9880	9300	2200	1142	989	1360	584	383	540	525
Silicon area, [mm^2]	0.0085	0.0082	0.0198	0.036	0.0050	0.057	0.0830	0.06	0.017	0.026

d) average $|V_{TH}|$

e) simulated

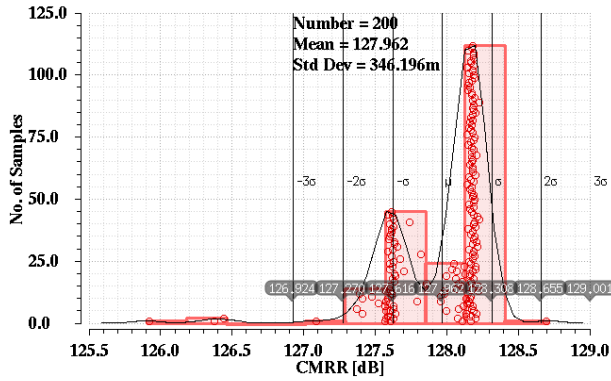


Fig. 9. Monte Carlo simulation of CMRR.

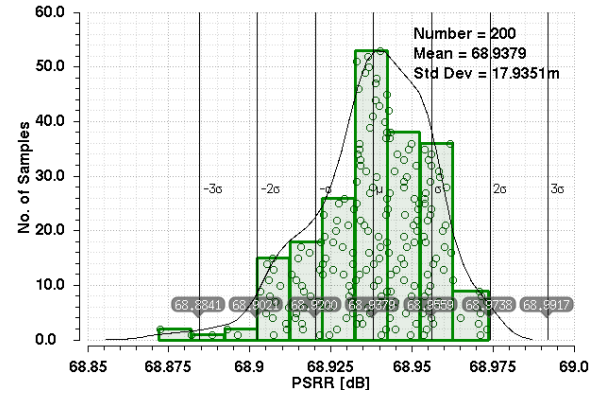


Fig. 10. Monte Carlo simulation of PSRR.

also exploits a nontailed differential amplifier at the input and can operate with identical V_{DD} .

Moreover, the proposed circuit outperforms all other ULV OTAs in terms of CMRR. It is worth pointing out that superior CMRR performance was obtained, despite using a nontailed architecture, suitable for ULV supply.

Comparing the input-referred noise of the OTAs in Table VI, we can notice large differences between different circuits. The differences are caused by the assumed biasing currents (GBW products) and assumed driving method. Typically, GD circuits

offer lower input noise but at the cost of reduced ICMR, especially in ULV environment.

The noise properties of the proposed OTA can be compared with the noise properties of the OTA in [10], which is supplied with similar V_{DD} and consumes similar power. Moreover, the total biasing current of the input differential stages of the two circuits is the same (10 nA). The input-referred noise of the proposed OTA is around two times lower than in [10]. This difference may be attributed to the fact that the OTA in [10] was realized using low- V_{TH} process with much lower g_{mb}/g_m ratio than the 0.18- μm process used in this design.

In order to compare small- and large-signal power effectiveness of the amplifiers in Table VI, the following well-known figures of merit (FOMs) have been adopted:

$$\text{FOM}_1 = \frac{\text{GBW} \cdot C_L}{I_{DD}} \quad (15)$$

$$\text{FOM}_2 = \frac{\text{SR}_{\text{aver}} \cdot C_L}{I_{DD}} \quad (16)$$

where SR_{aver} is the average SR and I_{DD} is the total supply current. The above FOMs, however, do not take into account the low-voltage capabilities of the compared OTAs, which can be better pointed out using the following FOMs:

$$\text{FOM}_3 = \frac{\text{GBW} \cdot C_L}{P_{\text{Diss}}} \quad (17)$$

$$\text{FOM}_4 = \frac{\text{SR}_{\text{aver}} \cdot C_L}{P_{\text{Diss}}} \quad (18)$$

where I_{DD} has been replaced with the dissipation power $P_{\text{diss}} = I_{DD} V_{DD}$.

Compared to the other ULV designs, the proposed OTA offers one of the best FOMs₁. Only the design in [11] offers better FOM₁, since its performance was optimized for one particular value of C_L , using the design procedure described in [23]. Note that both designs offer very similar FOM₃, since the OTA proposed in this article can operate with lower V_{DD} .

While comparing the large-signal behavior, expressed by FOM₂ and FOM₄, the proposed OTA outperforms all other solutions in Table VI.

Compared to the other sub- V_{TH} design in [16], the proposed OTA offers larger FOM_{1,3} and similar FOM_{2,4}. Even though the FOM_{1,2} in [16] could easily be improved with a nulling resistor in series with C_C , and its minimum SR is larger due to the more advanced class AB output stage, the design proposed here is simpler, more compact, and offers larger CMRR.

V. CONCLUSION

A simple solution for a ULV OTA was proposed. The design is based on a nontailed BD differential stage combined with an MMZCC technique and confirms good operation of such a configuration. Compared with other ULV designs the circuit described here offers very good power efficiency in terms of standard FOMs and much better CMRR, achieved in a compact and reliable structure.

APPENDIX

In order to compare the input-referred noise and offset of the proposed circuit with conventional BD pair, let us first consider the BD long-tail differential pair operating in a weak inversion region, as shown in Fig. 11.

A. Noise and Offset of the Reference BD Pair

Let us assume that the mean-square thermal ($\overline{i_t^2}$) and flicker ($\overline{i_{1/f}^2}$) drain noise currents of an MOS transistor operating in a weak inversion region are, respectively, given by

$$\overline{i_t^2} = 2nkTg_m \quad (A1)$$

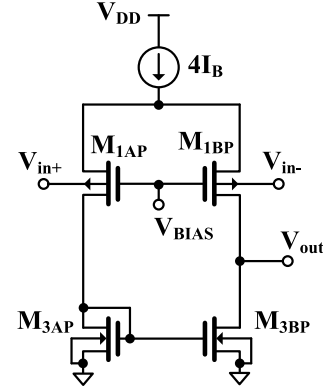


Fig. 11. BD differential pair.

$$\overline{i_{1/f}^2} = \frac{K \cdot g_m^2}{f C_{OX} WL} \quad (A2)$$

where n is the subthreshold slope factor, k is Boltzmann's constant, T is the absolute temperature, g_m is the transconductance, K is the flicker noise constant, and C_{OX} is the oxide capacitance per unit area. The input-referred thermal and flicker noise of the BD pair can be expressed as

$$\overline{v_{tp}^2} = 2 \frac{4n_p kT}{g_{m1p}} \cdot \frac{1}{\eta_p^2} \quad (A3)$$

$$\overline{v_{1/fp}^2} = \frac{1}{\eta_p^2} \cdot \frac{2}{f C_{OX}} \left[\frac{K_p}{W_{1p} L_{1p}} + \left(\frac{n_p}{n_n} \right)^2 \frac{K_n}{W_{3p} L_{3p}} \right] \quad (A4)$$

where $\eta_p = g_{mb1p}/g_{m1p}$. The input-referred noise voltage of the BD pair is increased $1/\eta_p$ times as compared to its GD counterpart, realized with the same transistors and biased with the same current.

The input-referred offset of the differential pair can be analyzed performing small-signal mismatch analysis [24]. For the benefit of compact, but sufficiently accurate expressions, it can be assumed that, in a weak inversion region, the random differences of threshold voltages dominate all other sources of mismatch, thus the other sources can be neglected [24], [25].

For a pair of closely spaced identical MOS transistors, the random differences of threshold voltages (ΔV_{TH}) can be described by a normal distribution with zero mean and a variance given by [26]

$$\sigma^2(\Delta V_{TH}) = \frac{A_{TH}^2}{WL} \quad (A5)$$

where A_{TH} is the process-dependent constant. Using the above model, the input-referred offset of the considered BD pair (standard deviation) may be expressed as

$$\sigma_p(V_{OS}) = \frac{1}{\eta_p} \sqrt{\frac{A_{THp}^2}{W_{1p} L_{1p}} + \left(\frac{n_p}{n_n} \right)^2 \frac{A_{THn}^2}{W_{3p} L_{3p}}} \quad (A6)$$

Thus, the input-referred offset depends on the transistor channel areas. Note that the input-referred offset of the BD pair is again increased by the factor of $1/\eta_p$ as compared to its GD counterpart.

$$\sigma(V_{OS}) = \frac{1}{\eta} \sqrt{\frac{A_{THp}^2}{4} \left(\frac{1}{W_1 L_1} + \frac{1}{W_2 L_2} \right) + \left(\frac{n_p}{n_n} \right)^2 \frac{A_{THn}^2}{4} \left(\frac{1}{W_3 L_3} + \frac{1}{W_4 L_4} \right)} \quad (A9)$$

B. Noise and Offset of the Nontailed Pair

The input-referred noise and offset of the OTA shown in Fig. 1 are both dominated by its input stage composed of transistors $M_{1A,B}$ – $M_{4A,B}$. Using the same noise and offset models as previously, and assuming further that $M_{1A,B} = M_{2A,B}$, which guarantees optimal noise performance [19], the input-referred thermal noise of this circuit is given by

$$\overline{v_t^2} = \frac{4n_p kT}{g_{m1}} \cdot \frac{1}{\eta^2}. \quad (A7)$$

Assuming $\eta = \eta_p$ and comparing (A3) with (A7), we obtain $\overline{v_t^2} = \overline{v_{tp}^2}$ if $g_{m1} = g_{m1p}/2$, i.e., when both circuits are biased with the same total current (note that this current in the input stage shown in Fig. 1 is divided into four branches instead of two).

The input-referred flicker noise of the nontailed pair is given by

$$\overline{v_{1/f}^2} = \frac{1}{\eta^2} \cdot \frac{2}{f C_{OX}} \left[\frac{K_p}{4} \left(\frac{1}{W_1 L_1} + \frac{1}{W_2 L_2} \right) + \left(\frac{n_p}{n_n} \right)^2 \frac{K_n}{4} \left(\frac{1}{W_3 L_3} + \frac{1}{W_4 L_4} \right) \right]. \quad (A8)$$

Note that, for $\eta = \eta_p$, the input-referred flicker noise of both circuits will be the same if $W_1 L_1 = W_2 L_2 = W_{1p} L_{1p}/2$ and $W_3 L_3 = W_4 L_4 = W_{3p} L_{3p}/2$. Thus, splitting each transistor of the BD pair shown in Fig. 11 into two identical twice smaller transistors, biased with a half of the current (thus maintaining the same total biasing current), ensures equal input noise, both thermal and flicker, for the two compared circuits.

It is worth noting that, in the general case, M_3 and M_4 shown in Fig. 1 do not need to be identical. In such a case, the flicker noise of the two compared circuits will be the same if $W_{3p} L_{3p} = 4W_3 L_3 / (W_3 L_3 + W_4 L_4)$.

The input-referred offset of the nontailed input stage is given by (A9), as shown at the top of this page.

The input-referred offsets of the nontailed and conventional BD pair are equal to each other, provided that the same conditions that ensure the equality of flicker noise are met.

It is also worth mentioning that, in many practical cases, $\eta > \eta_p$ due to the larger V_{SB} voltage of the input transistors at the operating point, which increases the g_{mb}/g_m ratio. Therefore, the input noise and offset of the nontailed pair biased with the same current and occupying the same area can be even lower than for a conventional BD pair.

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